

CURRICULUM VITAE

Cameron Wilhite

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Postdoctoral Scholar

Howard Hughes Medical Institute

University of California San Francisco

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EDUCATION

2019 – 2026 **Ph.D.**, Neuroscience, University of California San Francisco, San Francisco, CA

2016 – 2018 **M.S.**, Physiological Sciences, University of Arizona, Tucson, AZ

2010 – 2014 **B.S.**, Physiology, University of Arizona, Tucson, AZ

RESEARCH

2019 – 2026 **Ph.D. Student**, University of California San Francisco, San Francisco, CA

Primary Advisor: Massimo Scanziani; Secondary Advisor: Loren Frank

Dissertation: Modulation of turn-related activity in the superior colliculus by ongoing behavioral and cognitive dynamics ([pdf](#))

2016 – 2018 **M.S. Student**, University of Arizona, Tucson, AZ

Primary Advisor: Stephen Cowen; Secondary Advisor: Russell Witte

Research: Acoustoelectric and electrophysiological mapping of cardiac and neural activity

EMPLOYMENT

June 2026 – Present **Postdoctoral Scholar**, Howard Hughes Medical Institute

University of California San Francisco, San Francisco, CA

Advisor: Massimo Scanziani

2018 – 2019 **Research Specialist**, University of Arizona, Tucson, AZ

Experimental Ultrasound and Neural Imaging Laboratory

2016 – 2018 **Graduate Teaching Assistant**, University of Arizona, Tucson, AZ

Physiology Department

AWARDS AND FELLOWSHIPS

General awards

2019 – 2024 Discovery Fellow, UCSF Graduate Division

2020 – 2021 NIH Training Grant Student Award (T32), UCSF Neuroscience Graduate Program

Travel awards

2022 Graduate Student Award, Memory, Space, and Time Workshop, Tucson, AZ

2018 Infinity Conference Group Trainee Award, BRAIN Initiative Investigators Meeting, Rockville, MD

SCHOLARSHIP

Manuscripts

Wilhite C., Frank LM., Scanziani M., “Hippocampal representations of possible futures predict motor output in the superior colliculus,” under review, 2026. [bioRxiv link](#)

Wilhite C., Frank LM., Scanziani M., “Left-right alternation of superior colliculus activity with the stepping rhythm,” under review, 2026. [bioRxiv link](#)

Brenner JM.*, Ruediger S.*, **Wilhite C.***, Regalado JM., Senzai Y., Voskobiynyk Y., Paz JT., Scanziani M., Beltramo R., “Vision shapes neural maps of space through an ancient midbrain pathway,” under review, 2026. (*equal contribution) [bioRxiv link](#)

Alvarez A., Preston C., Trujillo T., **Wilhite C.**, Burton A., Vohnout S., and Witte RS., “In vivo acoustoelectric imaging for high-resolution visualization of cardiac electric spatiotemporal dynamics,” *Appl. Opt.* 59, 11292-11300, 2020.

Conference Presentations

Wilhite C., Frank LM., Scanziani M., “Directional alignment of turn-related activity in the superior colliculus with locomotor dynamics and hippocampal representations of future paths,” Lake Conference – Neural Coding and Dynamics, 2025 / Society for Neuroscience Meeting, 2025.

Wilhite C., Frank LM., Scanziani M., “Coordination between nonlocal hippocampal representations and the collicular orienting system,” Society for Neuroscience Meeting, 2024.

Wilhite C., Frank LM., Scanziani M., “A locomotor rhythm organizes directional firing of neurons in the superior colliculus,” Society for Neuroscience Meeting, 2023 / HHMI Science Meeting, 2023.

Wilhite C., Senzai Y., Scanziani M., “Poor spatial representation in primary visual cortex in the absence of visual input,” Society for Neuroscience Meeting, 2021.

Wilhite C., Witte RS., Cowen SL., “Parallel signal transfer through hippocampal CA2 region following perforant-path stimulation and internally generated dentate spikes,” Society for Neuroscience Meeting, 2019.

Wilhite C., Alvarez A., Burton A., Preston C., Gothard KM., Fuglevand AJ., Mustacich D., Cowen SL., Witte RS., “*In vivo* swine model for developing and validating acoustoelectric brain imaging of neuronal current,” BRAIN Initiative Meeting, 2019 / Society for Neuroscience Meeting, 2019.

Wilhite C., Witte RS., Cowen SL., “Peak activation of the CA2 sub-region of the hippocampus precedes peak activation of CA3 following perforant-path stimulation,” Society for Neuroscience Meeting, 2018.

Wilhite C., Burton A., Hill D., Bera T., Ingram P., Cowen SL., Witte RS., “Acoustoelectric brain imaging in anesthetized rats: towards noninvasive, real-time 4D electrical brain mapping,” BRAIN Initiative Meeting, 2018.

Departmental Talks

“Locomotion and mental simulation in the superior colliculus,” Ph.D. Thesis Seminar, University of California San Francisco, San Francisco, CA, 2026. ([recording](#))

“A locomotor rhythm in the superior colliculus phases hippocampal theta sweeps,” Research in Progress Seminar, University of California San Francisco, San Francisco, CA, 2024.

“A locomotor rhythm organizes turn-selective firing of neurons in the superior colliculus,” Research in Progress Seminar, University of California San Francisco, San Francisco, CA, 2023.

“Re-examining memory circuits: Role of hippocampal CA2 region in the initial processing of cortical input,” Neuroscience Data Blitz, University of Arizona, Tucson, AZ, 2019.

“Acoustoelectric brain imaging in the rat hippocampus: towards noninvasive, real-time electrical brain mapping,” Physiological Sciences, University of Arizona, Tucson, AZ, 2018.

TEACHING EXPERIENCE

2021 Teaching Assistant, University of California San Francisco, San Francisco, CA
Course: Cell Physiology

2016 – 2018 Teaching Assistant, University of Arizona, Tucson, AZ
Course: Human Anatomy and Physiology

SCIENCE LEADERSHIP

2023 – Present Journal Club Organizer and Lead, University of California San Francisco, San Francisco, CA
Current Topics in Physiology

SKILLS

Laboratory techniques

- Behavioral task design and real-time experimental control for rodent studies
- High-density *in vivo* electrophysiology using silicon probes
- Electrical stimulation for evoking neural currents *in vivo*
- Histology, staining, and imaging of neural tissue

Neural data analysis and computational approaches

- Programming: Python
- High-density mapping of evoked neural currents
- Phase-locking of neuronal activity to neural and behavioral oscillations
- Decoding of animal position from ensemble activity in the hippocampus
- Modeling functional coupling between brain areas using generalized linear models